Project Mechatronic Design 2026

Group Number 18

Tutor:

Bernard Hoppenreijs

Students:

Yaroslav Knyazkov (s3818187)

Fernando Nogueira Martins (s3814025)

Samson Mazurier (s3814017)

Noah En-Wel Lecomte (s3806553)

Zayan Ashraf Urothiyil (s3806545)

Mohammad Darrel Razan (s3806529)



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ign Problem

A precision mechanism with a single degree-of-freedom is to be designed for a laser weeding system. The mechanism is a stage which rotates a mirror about a pivot point in order to angle the laser. The laser must follow a specific motion profile to target weeds. A good design would support higher driving speeds by minimising the tracking error to allow for faster weeding.

Table 1 Requirements of the design

Requirements	Description
Accuracy	Maximum error of ±2mm from the target spot, which translates to an angular accuracy of ±0.931 mrad for the mirror.
Return time	Less than or equal to 30ms
Weeding time	780ms minus the return time
Motion Amplitude	±1 degree or ±17.5 mrad
Physical Setup	A vertical rotation axis for the mirror, with its centre point located between 50 mm and 150 mm from the base.
Non-functional requirements	Examples: cost, aesthetics, safety, compactness.

The provided hardware guides us on the constraints of the structure.

Table 2 The provided hardware and their specifications

Hardware	Type	Main Specification
Sensor	Magnetic encoder, RLS LM10	8192 counts / 2 mm
Actuator	Voice Coil Motor (VCM), AVM30-15	4.63 N continuous force
Actuator Amplifier	Custom built PWM voltage amplifier	-24 V to 24 V for duty-cycle -100% to 100%
Control Electronics	Custom electronics box with LAUNCHXL-F28379D LaunchPad™	C2000™ microcontroller, 2x200 MHz

Nominal Model

The parameters below are used to verify the nominal model of our system.

Table 3 Parameters for nominal model

Parameter	Symbol	Value
Force constant	k_f	7.35 N/A
Back-EMF constant	k_c	7.35 V/(m/s)
Coil resistance	R	10.24 Ω
Coil inductance	L	0.00282 H
Max Voltage	V_{max}	6.45 V
Lever arm	l_a	87.5 mm
Inertia (from CAD)	I	2.1e-4 kgm^2
Stiffness	c	24.334 Nm/rad
Period	T	0.75 s
VCM Max Angle	θ_{max}	±17.5 mrad

The limit of our nominal model is set with $V_{max} = Max\ Current * Coil\ Resistance = 6.45\ V$. The inertia is calculated in Fusion 360, and stiffness as $c = \frac{F_m l_A}{\theta_{max}}$, when derived from equation of motion at max angle θ_{max}.

The mechanism is modelled as a single-DOF rotational mass-spring system with the following equation:

$$I\ddot{\theta}(t) + c\theta(t) = F(t) * l_a$$

Which combines with VCM electrical dynamics, $V = Ri + Li + k_c\dot{x}$, as well as force current relationship, $F = k_f i$, and lever arm, $x = r\theta$, yielding the transfer function:

$$P(s) = \frac{\theta(s)}{V(s)} = \frac{-l_a k_f}{ILs^3 + IRs^2 + Lc + k_f k_c l_a^2 s + cR}$$

The peak actuator voltage is calculated with the electromechanical voltage equation:

$$V_{peak} = \frac{R}{k_f l_a}\left(I\ddot{\theta}_{max} + c\theta_{max}\right) - k_c l_a \dot{\theta}_{max}$$

The first term shows the resistive voltage that is needed to produce the mechanical torque, rotating the stage. The second term represents the back-EMF produced by the moving coil. We calculated $\dot{\theta}_{max} = \theta_{max} * \left(\frac{2\pi}{T}\right) = 0.147\ rad/s$, and $\ddot{\theta}_{max} = \theta_{max} * \left(\frac{2\pi}{T}\right)^2 = 1.23\ rad/s^2$. According to our parameters, $V_{peak} = 5.67\ V$, which is below the voltage limit of 6.45 V. This means our design is feasible.

Motivated Design of the Controller Structure

A PID controller with feedforward is chosen for this system. The flexure provides stiffness k, requiring integral action to eliminate steady-state error. The low damping requires derivative action to add phase lead and stabilize the loop. Since the reference trajectory is known in advance, feedforward based on the inverse plant model provides the bulk of the control effort, leaving feedback to correct only disturbances and model errors.

This two-degree-of-freedom structure separates tracking (feedforward) from stability and disturbance rejection (feedback), keeping error low without pushing gains to the stability limit. This approach is well-established for voice-coil servos with similar dynamics.

Stability Margins and Predicted Tracking Error

Plotting the nominal plant's open loop frequency response function, we found its crossover frequency, phase and gain margins. According to S&C textbook, the goal is to increase w_c to obtain a faster response, Phase margin for stability and damping, and Gain margin for robustness. The increasing of w_c is limited by delays due to discretisation and non-collocated parasitic eigenfrequencies, so further improvements to high frequency tracking and response would be done via feedforward. As for the phase margin, we considered using a lead filter, aiming for a margin of 30-60 deg.

To satisfy the requirements, we need to add gain that raises the magnitude to 0dB at selected crossover frequency of 200Hz. This would be in pair with a lead filter that improves the phase margin here. The graph below shows the time plot of our plant's error with a controller implementing the described changes.

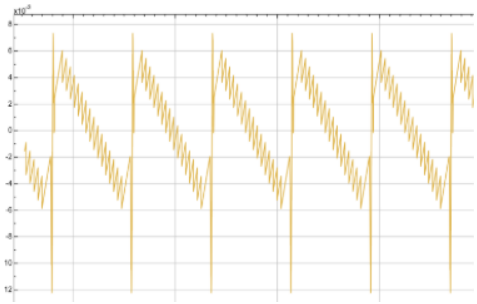


Figure 1: Error of nominal plant with implemented controler

Detailed Mechanical Design

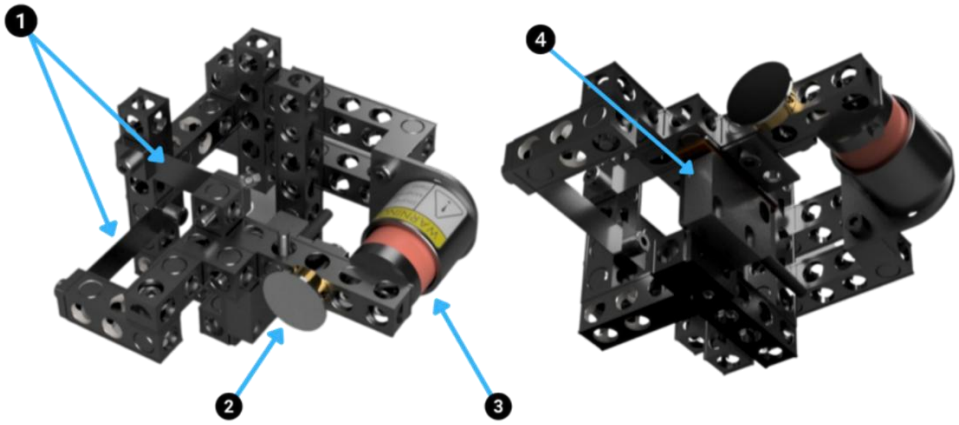


Figure 1 Labelled 3D CAD model of the structure

Our initial model was a cross-hinged design. However, we had a complication with the ordering of the springs and ended up with springs two time smaller as conceived. As the springs are still usable, a new design was made to adapt to the situation. The current system was optimised for a low moment of inertia. This structure is compact, and able to be shifted into different orientations with ease.
List of components:1 Leaf Springs; 2 mirror ; 3 Actuator: Voice coil motor, AVM30-15; 4 Sensor: Magnetic encoder, RLS LM10

Stiffness and Stress Analysis of the Leaf spring Flexure

Table 4 Flexure Geometry and Material Properties

Parameter	Value
Leaf spring thickness, t	0.4 mm
Leaf spring width, w	12.5 mm
Leaf spring length, L_1	22 mm
Leaf spring length, L_2	37.5 mm
Material	Stainless steel 1.4310
Young's modulus, E	190 GPa [1]
Proof strength, $R_{p0.2}$	195 MPa [2]
Maximum rotation, ϑ_{max}	±1° (±17.5 mrad)

Using the method displayed in “Mechanical Design for Control - part 1” on canvas and explained in the first integration lecture, the stiffness of each spring is calculated.
Both leaves deform simultaneously under the same rotation and act in parallel, so the equivalent rotational stiffness is $c = k_1 + k_2$

Stiffness calculation

With $I = 6.67 \times 10^{-14}$ m^4 and $EI = 1.33 \times 10^{-2}$ N·m^2:

$k_1 = 14.55$ Nm/rad, $k_2 = 9.78$ Nm/rad

Both leaves deform simultaneously under the same rotation and act in parallel, so the equivalent rotational stiffness is $c = k_1 + k_2$

c = 14.55 + 9.78 = **24.3 Nm/rad**

Which is below the maximum stiffness calculated in deliverable B of 26.51Nm/rad.

Stress calculation

$$\delta = \frac{My}{I}$$

The max stress at y=t/2, for a spring with length L:

$$\delta_{max} = \frac{Et}{L}\theta_{max} \qquad \frac{1}{SF} = \frac{E}{Rp_{0.2}} \cdot \frac{t}{L} \cdot \theta_{max}$$

Since for 190=~195, then

$$SF = \frac{L}{t \cdot 17.5}$$

So, the stress is independent of the width, proportional to the thickness and the angle, and inversely proportional to length. Our safety factors are 3.57 and 5.53, for the 25mm long spring, and the 37.5mm long spring respectively.

Extended Dynamic model and Modal Analysis

Modelling on SPACAR was done by placing nodes for the position of the springs and connecting them to the centre of mass using rigid elements. Important values like mass, inertia and centre of mass were taken from the CAD model and recreated as simply as possible on SPACAR.

The simulation yielded the different vibration modes and depicted the system behaviour for each. Then frequency bands for the different modes were defined to accurately obtain frequency response on a bode plot. Each of the 10 frequencies correspond to a certain deformation shape of the flexure in which only one is the required degree of freedom and the other modes are parasitic.

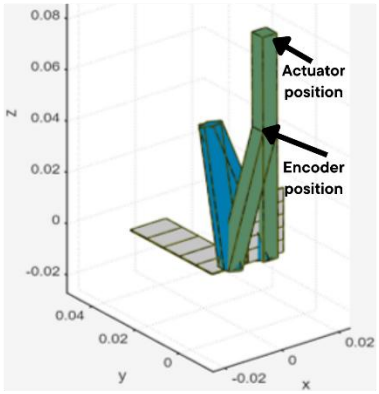


Figure 2: Spacar plant

The bode plots for all plant and the modal contributions are shown below. The first parasitic resonance occurs at mode 2 which is a non-intended degree of freedom. This occurs at 166.6 Hz or 1047 rad/s. It shows up as a sharp spike compared to surrounding and a rapid phase drop of approximately 180 degrees.

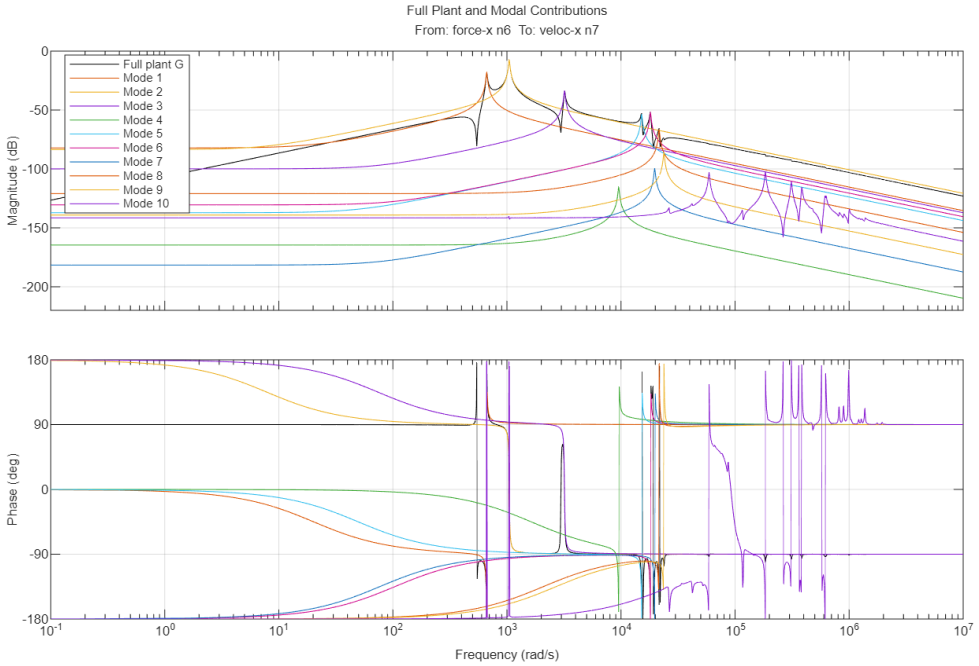


Figure 3 : Bode plot of the full plant and the separation of each mode up to 10

Regarding the inductance, $\omega_e = \frac{R}{L}$, tells us that with our modelled actuator transfer function, when we are below $\omega_e$the transfer function becomes purely resistive. However, when we are over this value, inductive term dominates and coil starts acting like a low pass filter (response rolls off at –20dB/decade).

By adding an actuator, the model gets scaled by the motor constant which changes the magnitude but not the shape for frequencies below ω_e.

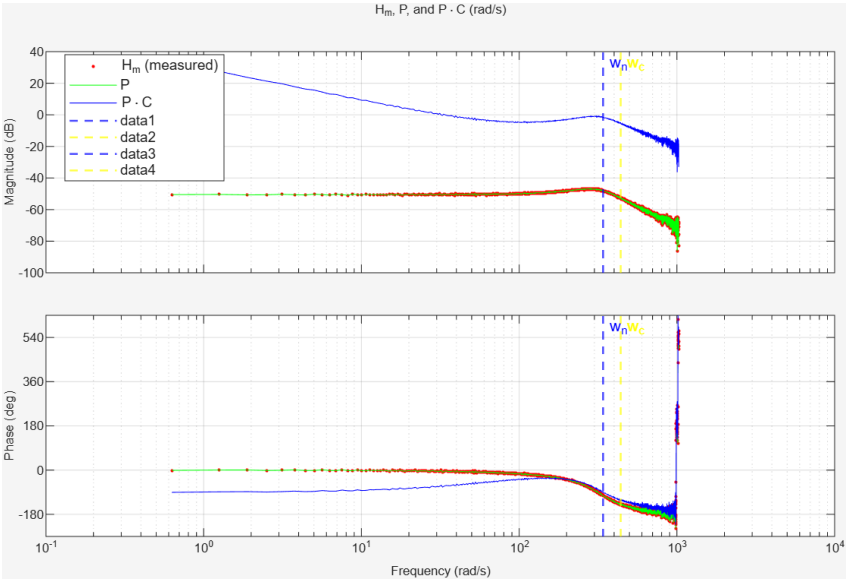


Figure 4

Discretisation Effects on Stability

We chose sampling of 0.0005, which is the maximum guaranteed sampling rate of the microprocessor. It was done to reduce effect of the effects of discretization. The best practice to choose sampling frequency 10 times

the crossover frequency. In our case we have magnitude of 10 quicker frequency.We used ZOH to discretize the plant and Tustin for the controller.

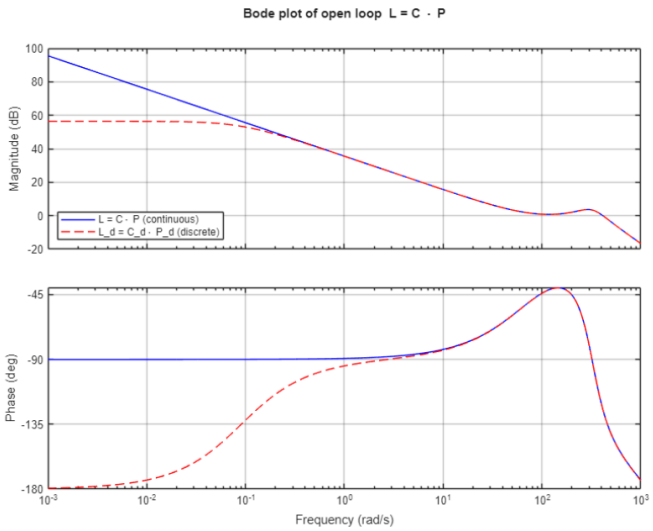


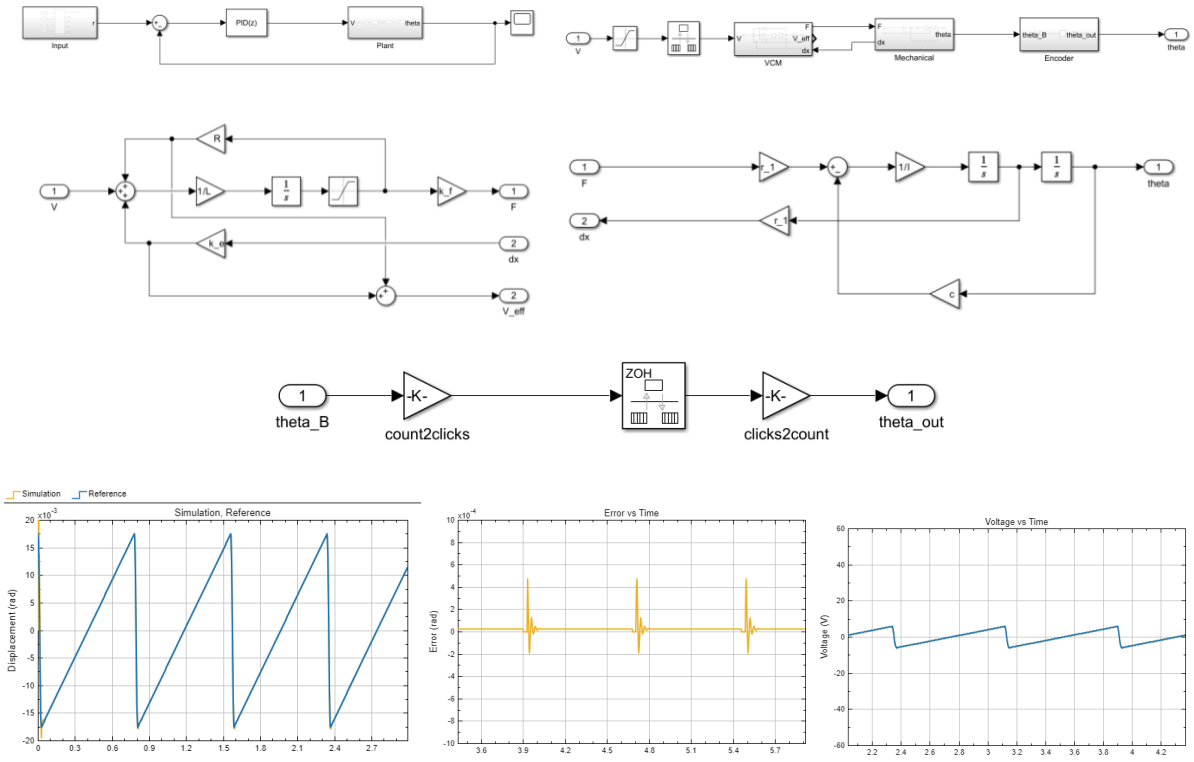
Figure 5 Bode plot comparing continuous and discrete controller plant

The stability of the system is PM = 50deg, GM = 20dB.

The plot shows comparison of discrete and continuous models. As can be seen discrete model starts flattening out as Tustin discrete has a pole at origin.

If margins degraded, we would propose: reducing Kp to reduce the crossover frequency; and reduce Kd or add phase lead to compensate for the lost phase at crossover. Optimizing the code to reduce computation and decreasing the sampling time, such that microprocessor can handle it.

Extended Closed-Loop Simulation



Design Trade-offs and Limits

Fundamental Design Limits:

- **Motor Parameters**: The VCM is limited to a maximum current (4A) and maximum voltage (60 V). The stroke of the motor restricts the lever arm length, which limits the max mirror angle.
- **Component Availability**: Limitation of components according to what was given for this project.
- **Microprocessor**: Guarantees a maximum computational frequency up to 2000Hz, which limits the sample rate.
- **Encoder**: has a resolution of 4096 counts/mm, limiting the precision of linear displacement measurement. This also contributes to quantization noise in the feedback loop.

Design Trade-offs:

- **Lever arm vs Angle**: Longer lever arm increases the mirror's max rotational angle for a given VCM stroke. We decided to prioritise getting the max ±17.5 mrad amplitude over maximising the theoretical angle.
- **Stiffness vs Moment of Inertia**: As the addition of more blocks would hold the structure better, it would also increase moment of inertia. We added a minimum number of blocks to prioritise the moment of inertia.
- **Crossover Frequency (ω_c) vs Parasitic Dynamics**: We need to improve tracking error and disturbance rejection by increasing the control bandwidth. However, maximising ω_c would excite the parasitic modes, which leads to instability. We pushed the ω_c to the maximum limit until the parasitic dynamics stopped further improvement.

Realised Mechanism

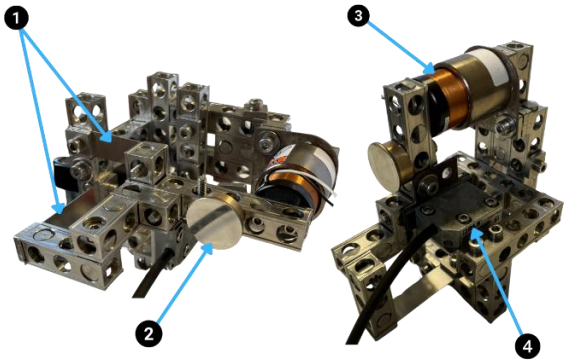


Figure 6 The real-life model of the mechanism. Correct orientation (Left).

The mechanism is mounted with a clamp on the base. The numbered components are consistent with the component list for the CAD model mentioned previously.

Static Gain Measurement

The method to obtain the static gain experimentally involved generating step voltage gains sent to the plant, waiting for the plateau to damp out, and repeat. Then the system response is analysed, a sample is taken 95% into the plateau for each step response.

For this example, 20 equal voltage step gains were inputted from –6.4512V to +6.4512V, each step corresponds to a period with a time per period of 10s. Using equation 5.66 from the DESIGN & CONTROL OF MECHATRONIC SYSTEMS lecture notes, zeta from back EMF and the natural frequency estimated with from Spacar, the 2% settling time is well below 10s, so 10s per period is a conservative choice.

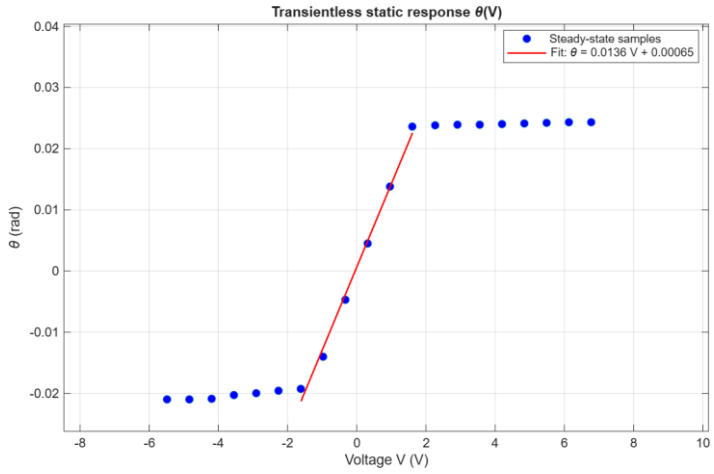


Figure 7 Experimental Static Gain

A clear linearity in the angle response to voltage is visible between –1.6128V and +1.6128V, while below and above these voltages it flats out. These flat sections come from the physical limits of the actuator where the stator hits the rotor.

Frequency Response Measurement

After completing the identification of the system, here's what we did and found:

We excited the system with two different types of signals - multi sine and chirp, with a frequency range of 0.5 to 10000hz. In order to ensure reliability, we also took note of the input force signal, derived from PWM, and output Theta. To create the frequency response function, we used the FFT command to generate Fourier coefficients of both the input and output signals and divided the output by the input at each frequency. We averaged this over multiple periods with the goal of reducing noise.

From this, we identified the inertia from high frequency approximation K/Jw^2; damping from the indicated phase of –90 degrees; stiffness from the low frequency plateau.

This allowed us to generate the coefficients to a plant that is more precise than our nominal plant. We used the characteristics of this plant to tune PID and feedforward to be better adapted to our plant, and study stability.

$$P_{manual} = \frac{1}{0.01598s^2 + 0.813s + 63.1}$$

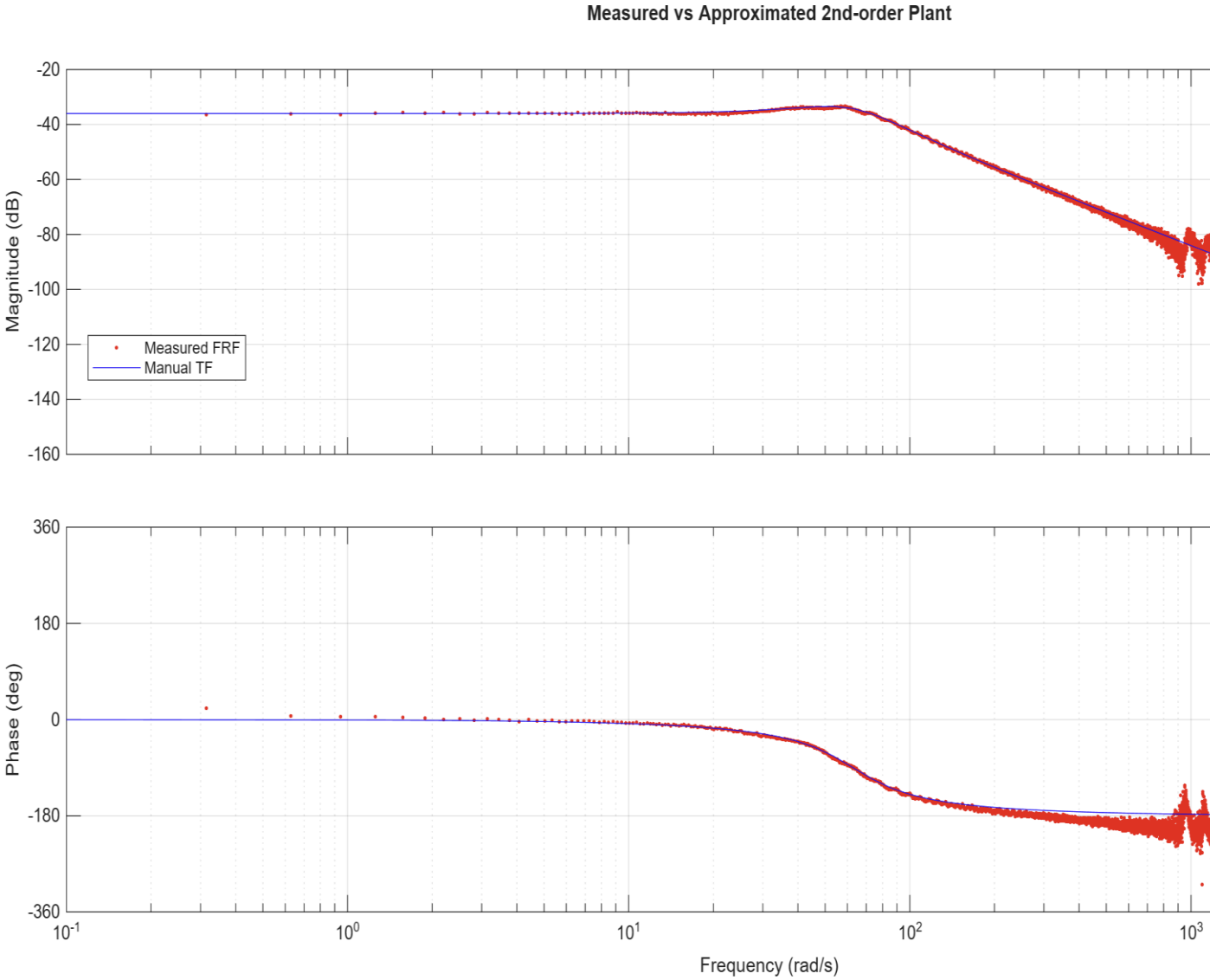


Figure 8 Approximated and collected frequency response of physical model

The figure shows the open loop frequency response functions of both the Nominal and experimentally obtained plants shown on the same plot. Both approximated transfer functions are unstable in the open loop, not crossing neither the 0dB line nor the –180deg line.

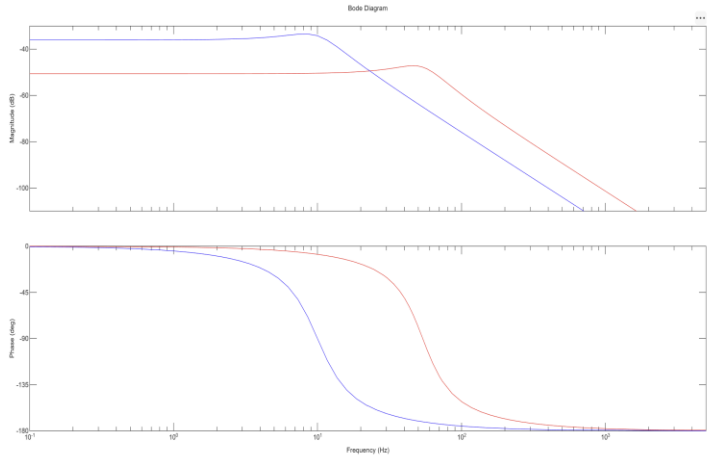


Figure 9 Bode plot of Nominal and Measured FRF

The model's primary eigenfrequency is at 334 rad/s versus at 64.8 rad/s for that of the measured FRF approximation, a ratio of 5.36 difference. There equally existed discrepancies in Inertia (High frequency) and stiffness (low frequency) of the same magnitude of difference. This scaling likely indicates a difference in calibration of the encoder between the two.

Experimental Stability Analysis

The next step was to use this experimentally obtained approximation of our plant to tune an ideal controller for it. During simulations, we noticed that feedforward only provided marginal improvements in error, therefore we decided to implement it only if necessary. As for the PID coefficients, we started with the values we determined in **Stability Margins and Predicted Tracking Error,** and varied them to optimise margins.

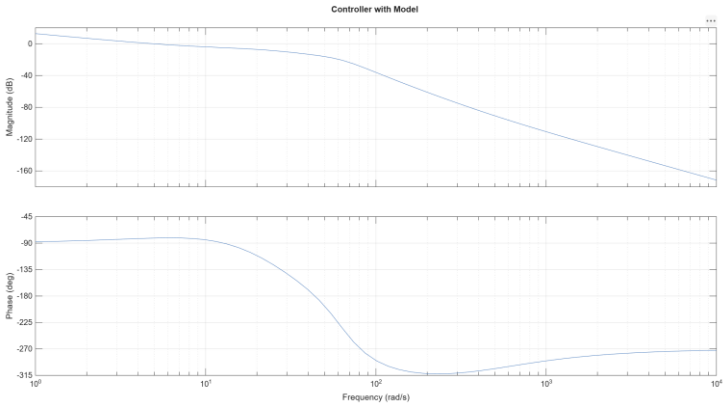


Figure 10 Bode diagram of optimised controller and plant

The implementation of this controller brings the gain margin up to 6dB, the Phase margin to 99 degrees, and provides a crossover frequency of 44 rad/s.

Experimental Tracking Results

In the reference function, we changed the return to be 10ms sinusoidal 20ms step and 750ms weeding time. To verify the validity of our model, experimental values were taken and compared with the reference function and limits. This is shown on the graphs.

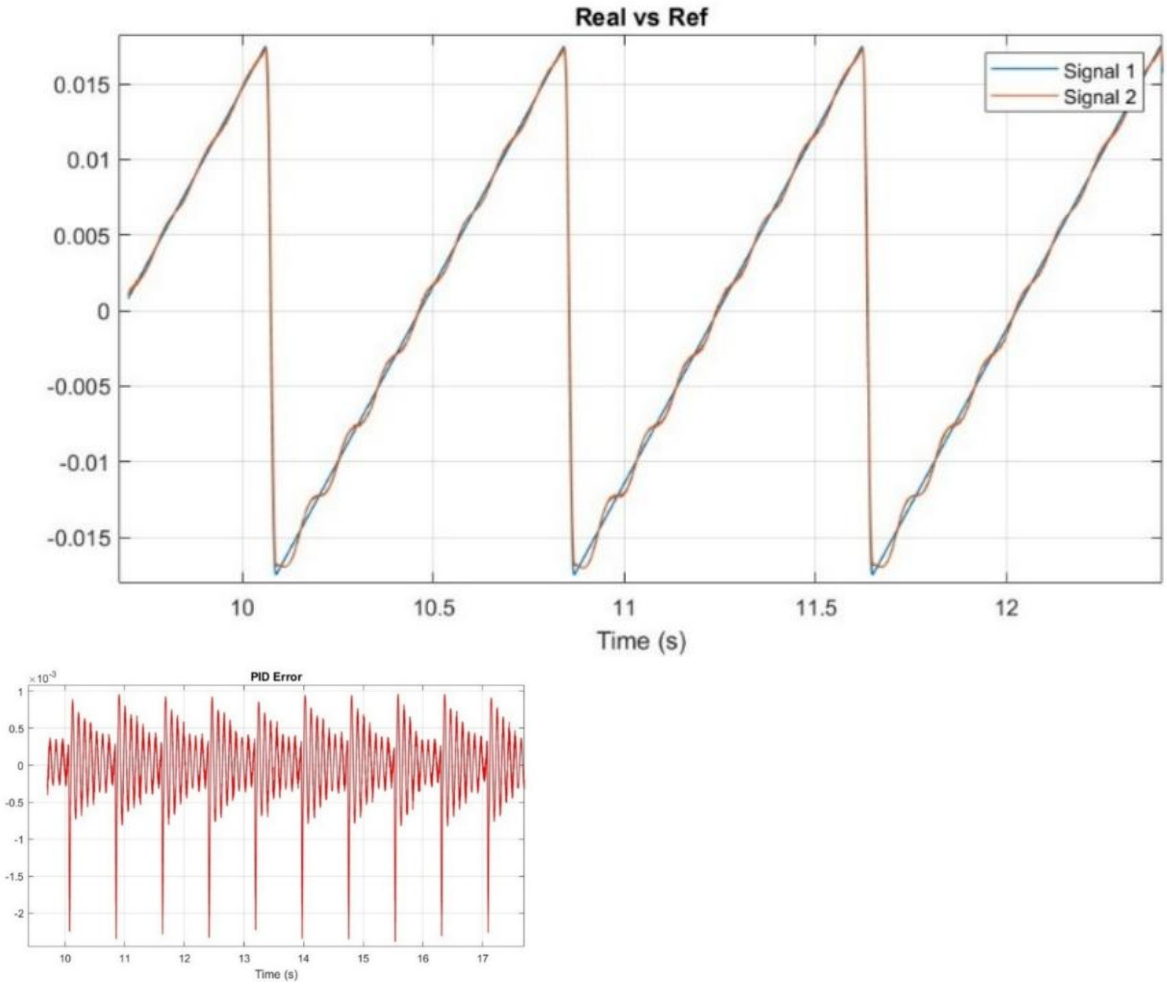


Figure 11/ Figure 12

Our real-life model matches up very well with the reference as shown in the graph above, signal 2 represents the motion of the real system, while signal 1 shows the reference trajectory. The PID error must be below ±0.931mrad, and our values show that they remain around ±0.9mrad at the peaks. The lower peaks are shown to be much higher in magnitude, but this is due to the return of the mechanism.

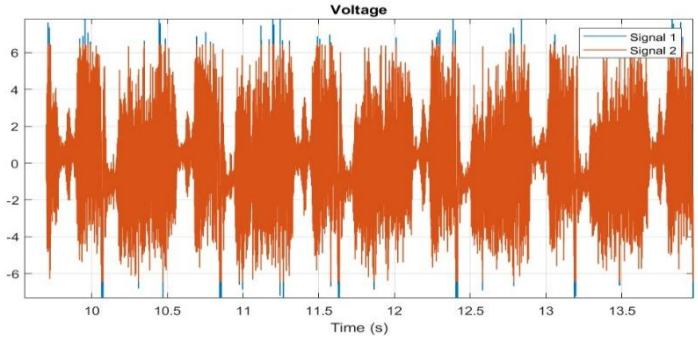
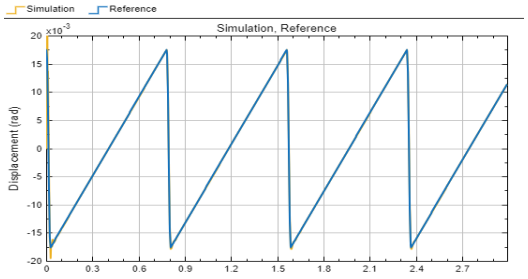
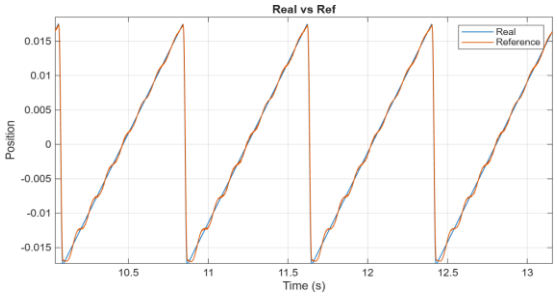


Figure 13

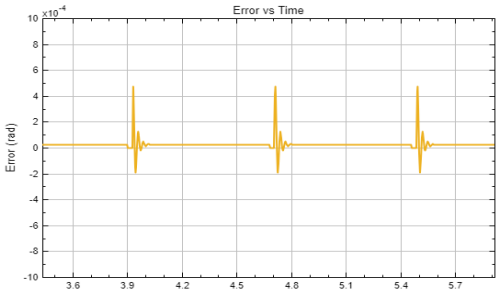
The voltage graph shows signal 1 as the actuator voltage, and signal 2 is the filtered voltage (below 6.45V). The graph shows that the voltage stays below the 6.45V limit.

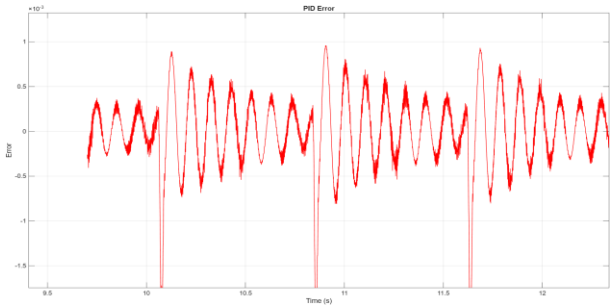
Simulation vs. Experiment Comparison



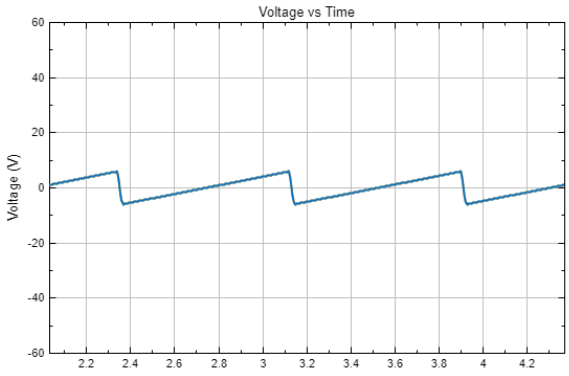


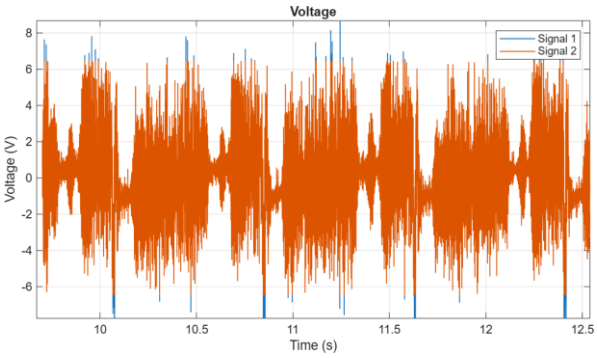
Overall shape matches each other. Experimental results contain noise likely due to parasitic resonance.





The real system has a persistent oscillation riding on top of the tracking error that isn't in the simulation at all. This looks consistent with exactly the parasitic resonance issue, which the linear simulation either doesn't include or has heavily damped out.





Performance Verification

Table 5 Pass/Fail requirements table

Requirements	Expected	Result	Description
Accuracy	Max error ±0.931 mrad	Pass. Max error ±0.930 mrad.	Error was within required accuracy
Return time	≤ 30ms	Pass. 24ms	Return time was within required time
Motion Profile	Amplitude ±1° /17.5 mrad	Pass. 17.5mrad	Full motion range obtained
Physical Setup	Mirror rotation on vertical axis and between 50 and 150 mm from base	Pass. Mirror rotation on vertical axis, 53mm from base	

Non-functional requirements		Cost, Aesthetics, Safety, Compactness.	

Evaluation and Recommendations

The final design meets its primary requirement of trajectory-accuracy compliance, which remained the fixed objective around which every other decision was made. Within the constraints available, the mechanism achieves the maximum arm angle the application requires, the natural frequency was maximised to reduce reference velocity and acceleration error, and the controller was pushed to a high crossover frequency for good closed-loop response. These are the design's clear strengths.

Much of the process, however, was governed by hard limits and unavoidable trade-offs. The motor was predetermined, fixing the maximum current and therefore the maximum force, which in turn set the flexure stiffness ceiling and the minimum arm length; the stroke capped the arm length from the other side. The microprocessor limited the sampling rate to 2000 Hz, and the encoder resolution of 4096 counts/mm bounded measurement precision. A restricted block count ruled out a multi-spring arrangement and forced a compromise between structural stiffness and moment of inertia, which we resolved by prioritising low inertia. Each performance gain came at a cost: a longer arm improves angle compliance but lowers the maximum angle, a higher natural frequency reduces tracking error but excites spring parasitic dynamics, and a higher crossover frequency improves response but is ultimately blocked by those same parasitic frequencies.

The most significant weakness was ordering the wrong spring lengths, which left the flexures too stiff. This forced a longer arm, raised the moment of inertia, and lowered the achievable natural frequency — degrading the very property the design sought to maximise — while spring parasitic frequencies became the binding limit on crossover frequency.

To improve the system, the single most effective change would be correcting the spring specification, recovering the lost natural frequency and tracking performance. Beyond that, a larger block budget to suppress parasitic frequencies, a faster processor to raise the sampling rate and recover phase margin, and a higher-resolution encoder would each yield further gains. Overall, the design is sound and satisfies its accuracy specification; its performance was limited less by fundamental physics than by the spring-length error, whose correction offers the clearest path to a better system.

During the preparation of this work the author used Deepseek in order to brainstorm structure, check for grammar mistakes and improve sentence clarity. After using this tool/service the author reviewed and edited the content as needed and takes full responsibility for the content of the work.

Reference List

[1] Young modulus of the leafspring https://www.alloywire.com/alloys/stainless-steel-1-4310/

[2]Yeald point of the leafspring https://www.kippcom.nl/en/1.4310-stainless-steel